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# Identification of geographical origins of soybean pastes using headspace gas chromatography-mass spectrometry by selecting sample-descriptive components with an Incremental Association Markov Blanket

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ABSTRACT

The identification of geographical origins of soybean pastes using headspace gas chromatography-mass spectrometry was attempted in this study. Since soybean paste was odor-rich, 36 components were identified in the imported and domestic soybean samples. *t*-Test, variable importance in projection (VIP), and Incremental Association Markov Blanket (IAMB) were employed to select proper components that could effectively discriminate the two sample groups. The discrimination accuracies were below 87.3 % when all 36 components were fed for either LDA, *k*-NN, or SVM. When the five *t*-test-selected components or six VIP score-selected components were employed, the accuracies improved to 95.2–96.2 %. The IAMB selected three different components were 3-methylbutanal, 4-methylnonane, and 2,3-pentanedione, and the correlations among their peak areas were not significant. This suggests that these three components were independently relevant for the discrimination. The accuracy obtained using these three components was superior, 97.7 %, as undescriptive and/or redundant components for the discrimination were excluded.

## 1. Introduction

Soybean paste (Shukla et al., 2014) is a popular condiment produced by fermentation of soybeans in brine and is consumed with rice, meat, and vegetables in East Asia. Soybean paste contains diverse nutrients such as proteins, amino acids, flavonoids, vitamins, and minerals (An et al., 2016; Jeong et al., 1998), and it is thereby regarded as a healthy food. Since import and export of foods including soybean paste have grown rapidly (aTFIS, 2021), and also the prices of soybean pastes vary depending on geographical origins of raw soybean used for the production (KAMIS), subsequent authentication of the product is necessary to prevent a potential fraud on the origin of soybean paste that is intended to gain an illegal profit. For the authentication, analytical tools enabling fast and non-destructive analysis are preferred to promptly handle large numbers of samples in market.

To saturate the demand, our group previously demonstrated the strategy of combining laser-induced breakdown spectroscopy (LIBS) and near-infrared (NIR) spectroscopy for the discrimination of soybean paste samples according to geographical origin (Jeong et al., 2023). When the areas of C ($A_C$) and Ca ($A_{Ca}$) peaks in the LIBS spectra were adopted as the inputs for linear discriminant analysis (LDA), the accuracy was 95.4 %. Alternatively, the use of the principal component (PC) scores obtained from the two-trace two-dimensional (2T2D) correlation NIR spectra resulted in a similar accuracy, 95.0 %. Interestingly, when the ratios of $A_{Ca}/A_C$ and the 2nd PC scores of the samples were used together, the accuracy improved to 99.6 %. Since the compositions of both elements and organic components were indicative of the samples' geographical origins, the combination of two complementary spectroscopic signatures was cooperative to enhance the accuracy.

Meanwhile, there are two weak points in the previous analytical scheme, although it yielded the superior discrimination accuracy. First, the discrimination by NIR spectroscopy relies on statistical pattern recognition of the measured spectral features, which are broad and highly overlapped owing to the characteristics of overtone and combination NIR vibrations. Due to the degraded component-specificity of NIR feature, explanation on origin of achieved result such as which

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components contribute to a given discriminant analysis is rather vague and less persuasive. Second, the use of two separate analytical methods requires more labor and is potentially inefficient. Therefore, a single analytical method enabling of providing discrete and specific component information useful for the discrimination is desirable and headspace gas chromatography-mass spectrometry (HS-GC-MS) has been adopted as a candidate in this study.

Chromatography can be used to classify samples with complex compositions such as agricultural products (Ganneru et al., 2020; Oliveira et al., 2016), foods (Qi et al., 2021; Zhao et al., 2016), and liquors (Vaclavik et al., 2011) into categories by separation of involved components. The use of entire separated components for classification is a common practice, since the concentrations of components are characteristic of classifying sample groups. Meanwhile, selection of certain components more discriminant of the sample group (class) could be an alternative to improve classification accuracy. For this purpose, statistical analysis (such as *t*-test and ANOVA) and variable importance in projection (VIP) scores based on partial least squares-discriminant analysis (PLS-DA) have been adopted to select descriptive components in chromatographic data. For example, red wines have been discriminated according to variety (Vaclavik et al., 2011), *Artemisia annua* L. in different growth stages was investigated (Ma et al., 2008), cocoa nibs from distinct origins were differentiated (Oliveira et al., 2016), *Carica papaya* Linn. was characterized by ripening practice (Ganneru et al., 2020), and volatile compounds were identified in royal jellies produced with different feeds (Qi et al., 2021).

For chromatography-based discrimination of geographical origins of soybean pastes, HS-GC-MS data obtained from the soybean paste samples produced using imported and domestic soybeans (simply referred to as imported and domestic soybean paste samples, respectively) were analyzed. For variable (component) selection from the chromatographic data, Incremental Association Markov Blanket (IAMB) has been readily adopted in this study. IAMB initially implements a grow phase to include all possible variables that are dependent of a target analysis and then a shrink phase to remove redundant variables. Although IAMB was utilized to select the descriptive variables in near-infrared (NIR) spectroscopic data possessing high multiple collinearities among the variables (Chang et al., 2018), it seldom has been adopted to encounter chromatographic data with discrete variables (peak areas).

First, peak areas of all separated components were used as the inputs for LDA, *k*-nearest neighbor (*k*-NN), and support vector machine (SVM) to discriminate the samples into the two groups. Second, the peak areas of components selected by *t*-test, VIP score, and IAMB were fed into discriminant analyses, and the resulting accuracies were compared with those using the peak areas of all separated components. Third, the components selected by *t*-test, VIP score, and IAMB were compared, and their contributions to the discrimination were discussed based on the corresponding loadings obtained by principal component analysis (PCA).

## 2. Materials and methods

### 2.1. Collection of soybean paste samples

For this study, 12 imported and 50 domestic commercial soybean paste samples were provided by the National Agricultural Products Quality Management Service (NAQS) in Gimcheon, Korea. The NAQS is an official governmental organization in Korea that verifies the country-origin of agricultural products. All the samples were collected from May to July in 2021, and their shelf lives were generally within 1 or 2 years. The domestic samples were directly provided by the corresponding manufacturers to NAQS, and the imported samples were purchased from local markets. All the employed samples were produced by the traditional fermentation method including only soybeans and salts. The detailed compositions of each sample are summarized in the supplementary material (Table S1). Among a total of 12 imported samples, 3 and 7 samples were produced from American and Chinese soybeans, respectively, and the remaining 2 samples used mixed American/Canadian/Chinese soybeans. It is worthy to note that the above 62 samples did not contain any additives. Further, as external validation samples, 5 imported and 10 domestic commercial soybean paste samples were separately prepared. Among these imported and domestic samples, 3 and 6 samples, respectively, contained additives (barley, monosodium glutamate, potassium sorbate, and etc.). The detailed descriptions on each sample are also presented in Table S1.

The samples (ranging from 500 to 900 g) were housed in either sealed glass or plastic containers, and immediately stored in a refrigerator to prevent further deterioration. At the same period, 100 g of soybean paste was taken from each sample and freeze-dried. Finally, each of the freeze-dried samples was separately ground by a grinder and used for subsequent HS-GC-MS measurements. In overall, the sample collection and pretreatment were accomplished over the three-month period.

### 2.2. HS-GC-MS analysis of soybean paste samples

A sample of each soybean paste powder (0.3 g) was transferred into a 20 mL glass vial (diameter: 23 mm, height: 75 mm). Then, 1 μL of 1000 mg/L 2-methoxy-3-methyl pyrazine (TCI Chemical Co., Tokyo, Japan) in distilled water was added as an internal standard. GC-MS analysis was performed by extracting the gas phase of the upper layer (headspace) in the vial. The vial containing the soybean paste samples was incubated for 10 min at 70℃ in an auto-sampler (Agilent, 7697A headspace sampler). The gas phase (1 mL) of the upper layer of the vial was transferred to a 6890 N GC-MS (Agilent, GC-6890 N coupled with GC-MS 5975B) located at CNU Chemistry Core Facility (Daejeon, South Korea) under the following conditions: loop temperature of 80℃, transfer line temperature of 100℃, and injection duration of 0.4 min. Helium (99.999 %) was used as a carrier gas with a flow rate of 1 mL•min$^{-1}$. A DB-WAX column (30 m, 0.25 mm I.D., 0.50 μm film thickness, Agilent, USA) was used for separation with the following parameters: injector temperature of 200℃, sample injection volume of 1 mL, and split ratio of 20:1. For temperature gradient analysis, the oven temperature was initially held at 40 ℃ for 5 min, increased to 200 ℃ at a heating rate of 10 ℃•min$^{-1}$, and held for 5 min. Ion source and quadrupole temperatures were 230 ℃ and 150 ℃, respectively. The mass spectrometer was operated in positive electron ionization (EI) mode at 70 eV. Data acquisition was executed with a total ion chromatogram (TIC) mode, and the mass range was selected from $m/z$ 33 to 500.

Agilent MassHunter Data Acquisition Software was used for peak detection and peak area calculation of the internal standard and all the identified compounds in all the raw data obtained by HS-GC-MS. The NIST Mass Spectral Search 2.3 Program (NIST, Washington, DC, USA) was used to identify the volatile compounds. The retention index values of the identified compounds were calculated by comparing the retention times of the identified compounds with those of C7-C40 saturated alkanes (Supelco, Bellefonte, PA, USA). The top 20 candidates with the highest probability from the search results for each peak were manually inspected by comparing their mass spectra to the experimental mass spectrum and comparing their retention index values to the experimental retention index value. The compound that was deemed most similar was selected as the correct compound. The relative peak areas of the volatile compounds compared to the area of the internal standard were used for subsequent discriminant analysis.

### 2.3. Variable (peak) selection and discriminant analysis

*t*-Test was performed to analyze statistical significance of differences in peak areas of separated components between the imported and domestic samples. Initially, Levene's test was employed to evaluate equality of variances for samples in the two groups. When the resulting *p*-value was greater than 0.05, the difference in variances in the two groups was not significant, and two-sample *t*-test was performed. In the

other case ($p$-value less than 0.05), Welch's $t$-test was executed. Python (version 3.8.5, Python Software Foundation) with Scipy package (Virtanen et al., 2020) was used to conduct the two-sample $t$-test, Levene's test, and Welch's $t$-test.

VIP scores (Andersen & Bro, 2010) are the loading weights for features obtained by partial least squares-discriminant analysis (PLS-DA) and indicate the contribution of each variable to targeting analysis. VIP scores were calculated for feature $j$ by Equation (1),

$$VIP_j = \sqrt{\frac{J \bullet \sum_{f=1}^{F} w_{jf}^2 SS_f}{SS_y F}} \quad (1)$$

where $w_{if}$ is a PLS-DA weight of feature $j$ on the latent variable (LV, or loading) $f$. $J$ and $F$ are total numbers of features and LVs, respectively. $SS_f$ is the percentage of the explained residual sum of squares, and $SS_y$ is the total percentage of the explained residual sum of squares. VIP scores greater than 1 are generally regarded as significant contributors (Shetty et al., 2012).

IAMB (Yaramakala & Margaritis, 2005) consists of grow and shrink phases for discovering the Markov blanket (MB). In the grow phase, features (variables) determined to be conditionally dependent on target analysis (discrimination) by confidence interval (CI) test are included in the MB until there are no additional features to add. In the subsequent shrink phase, false positives among the selected features are eliminated to minimize the number of features in MB. In execution of IAMB, CI is determined using the $G^2$ test, the $\chi^2$ test, or Fisher's $z$-test (Peña et al., 2007). $G^2$ and $\chi^2$ tests were used to deal with categorical features, and Fisher's $z$-test was used to handle continuous features. Therefore, Fisher's $z$-test was adopted in this study with significance level (α) for CI of 0.01. The IAMB was executed using MATLAB, version R2020b (MathWorks Inc., Natick, MA, USA).

For LDA, singular value decomposition (SVD) was adopted as a solver parameter. Euclidean distance was used as the distance metric for $k$-NN, and a linear kernel was adopted for SVM. Two-level cross validation (CV) loops (Guo et al., 2017; Varmuza et al., 2014) were used to evaluate discrimination performance of each classifier using 4 (outer)- and 3 (inner)-fold CV. Since the numbers of imported and domestic samples in the internal validation set (12 and 50 samples, respectively) were unbalanced, all the samples were stratified for sampling in CV. In the inner loop, an optimal number of neighbors of $k$-NN and cost constant of SVM were determined through grid search using GridSearchCV. Resulting F1 scores at each grid search were examined for determination of related parameters. Therefore, the numbers of neighbors ($k$-NN) and cost constant (SVM) determined in each cross validation were somewhat different to each other. As reported earlier (Guo et al., 2017), the use of optimal parameters for each cross-validation step was recommended to secure accuracy of target discriminant analysis. In the outer loop, discrimination performances of built models were assessed by predicting the test samples. This procedure was repeated 100 times for CV using a randomly shuffled dataset in each case. The resulting mean accuracy, specificity, and sensitivity were calculated. Finally, using the built models in each case, the classes of external validation samples (5 imported and 10 domestic samples) were determined and resulting discrimination performances were assessed. PLS-DA, PCA, LDA, $k$-NN, SVM, and GridSearchCV were performed using Python with the sklearn package (Pedregosa et al., 2011).

## 3. Results and discussion

### 3.1. Chromatograms of imported and domestic soybean paste samples

The chromatograms measured from the imported (red) and domestic (black) soybean paste samples were separately averaged, and the average chromatograms were examined as shown in Fig. 1. The strongest peak corresponding to acetic acid is observed at 15.1 min, and the second strongest peak of ethanol is positioned at 5.1 min. For detailed examination, the remaining peaks with low intensities are magnified in the inset. Among them, the 11 peaks are labeled with the corresponding components. As seen, many diverse peaks of volatile components in the soybean paste samples appear over retention times (RTs) from 2.0 to 19.0 min. Thirty-six components were simultaneously identified in both chromatograms. No component present only in one chromatogram was found. Table S2 (refer to the supplementary information) summarizes all

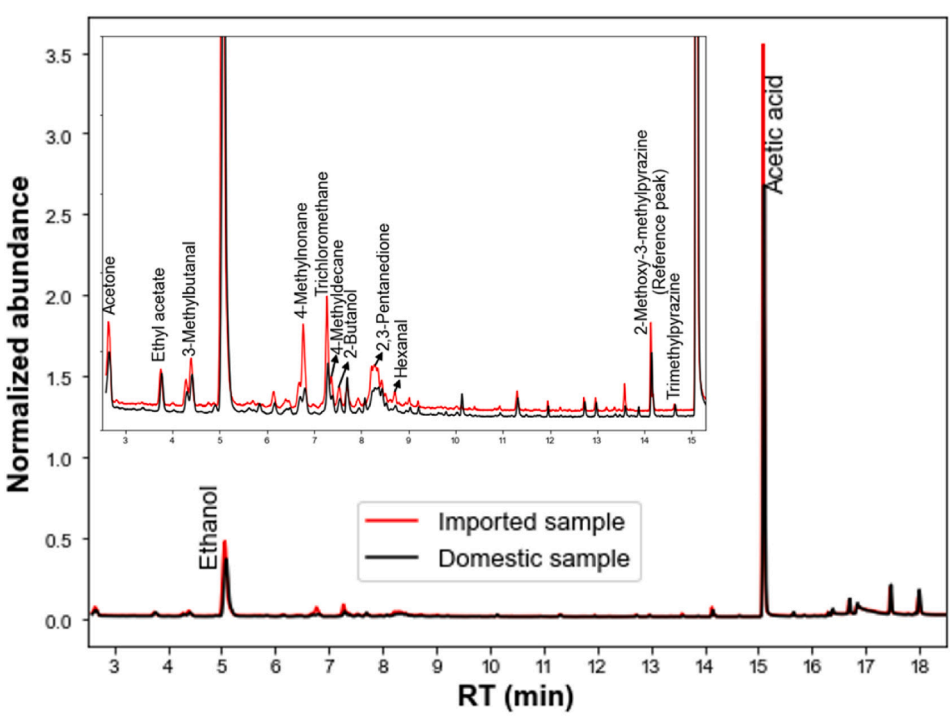


**Fig. 1.** Average chromatograms of the imported (red) and domestic (black) soybean paste samples. The inset highlights the peaks with low intensities. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

36 identified components with corresponding RTs, normalized peak areas relative to the peak area of 2-methoxy-3-methyl-pyrazine (internal reference), experimental retention index values, and reference retention index values. Less than 3 % differences were observed between experimental and reference retention index values for all of the identified compounds.

The current peak identifications were primarily achieved by comparing the acquired mass spectra with the mass spectra and the calculated retention index values with those available in the NIST database. However, in order to enhance the confidence of identification, additional information such as high resolution mass spectrum data and a reference standard would be necessary as previously suggested (Schymanski et al., 2014). With incorporating these supplementary resources in this study, the accuracy and reliability of peak identification could be improved. As seen in the Fig. 1, the relative abundance of all components differed between the imported and domestic samples and the dissimilar composition could be valuable for discrimination.

Further, to evaluate the reproducibility of retention times in the GC-MS measurement, the chromatograms obtained from three domestic (#2, #3, and #6) and three imported (#20, #31, and #42) soybean paste samples were chosen, and the corresponding retention times of three major components (acetic acid, ethanol, and butanoic acid) were analyzed. The corresponding chromatograms are shown in the supplementary information (Figure S1). The relative standard deviations (RSDs) of the retention times for acetic acid, ethanol, and butanoic acid were 0.03 %, 0.22 %, and 0.06 %, respectively. As provided, the reproducibility of employed GC-MS measurement was superior.

### 3.2. Discriminant analysis using peak areas of components selected by t-test

Using the 36 peak areas of all the separated components as the inputs (variables), discrimination of imported and domestic samples in the internal validation set was performed using three popular classifiers: LDA, *k*-NN, and SVM. Table 1 shows the average accuracies, sensitivities, and specificities obtained by 100-time cross-validation using the three classifiers. The corresponding standard deviations are also shown. Among the three cases, SVM provided the highest accuracy (87.3 %). To visually examine the discriminability provided by the 36 peak areas in a reduced dimension, PCA was performed using these peak areas, and the subsequent PC scores were investigated. Fig. 2 (a) shows the distribution of the 1st and 2nd PC scores corresponding to the 36 peak areas. The numbers in parentheses are the percentages of variation explained by the PC relative to the total variation. As seen, the PC scores of imported and domestic samples broadly overlapped, limiting differentiation of the two groups. The use of all separated components was ineffective for discrimination.

**Table 1**
Accuracies, specificities, and sensitivities resulting in the discrimination of imported and domestic soybean paste samples in the internal validation set when peak areas of all 36 separated components or selected components among them by *t*-test, VIP score, and IAMB are separately employed for LDA, *k*-NN, and SVM.

| Input | Classifier | Accuracy | Specificity | Sensitivity |
|---|---|---|---|---|
| **Peak areas of all 36 separated components** | LDA | 79.6 ± 4.6 % | 75.2 ± 10.1 % | 80.6 ± 5.2 % |
| | *k*-NN | 75.1 ± 2.9 % | 6.25 ± 5.7 % | 91.6 ± 3.4 % |
| | SVM | 87.3 ± 3.1 % | 68.9 ± 10.9 % | 91.6 ± 3.1 % |
| **Peak areas of 5 components selected by *t*-test** | LDA | 95.2 ± 1.3 % | 90.2 ± 4.5 % | 96.4 ± 1.2 % |
| | *k*-NN | 96.2 ± 1.7 % | 92.3 ± 6.7 % | 97.2 ± 1.4 % |
| | SVM | 95.2 ± 1.6 % | 88.1 ± 5.2 % | 96.9 ± 1.4 % |
| **Peak areas of 6 components selected by VIP score** | LDA | 94.8 ± 1.3 % | 90.5 ± 5.3 % | 95.8 ± 0.9 % |
| | *k*-NN | 94.7 ± 1.8 % | 91.1 ± 7.6 % | 95.6 ± 1.4 % |
| | SVM | 95.0 ± 1.6 % | 88.2 ± 5.4 % | 96.6 ± 1.7 % |
| **Peak areas of 3 components selected by IAMB** | LDA | 97.3 ± 0.9 % | 93.7 ± 3.8 % | 98.2 ± 0.6 % |
| | *k*-NN | 97.3 ± 1.3 % | 96.2 ± 6.4 % | 97.9 ± 0.4 % |
| | SVM | 97.7 ± 0.8 % | 97.8 ± 3.9 % | 98.0 ± 0.3 % |

Fig. 2 (b) presents the loadings used to generate the scores. Only 5 components with the greatest loading weights (equivalent to the longest lines from the point of zero weight) are shown. The weights are highest for ethanol and acetic acid, as expected due to their highest peak areas. Low weights were observed for trimethyl pyrazine, ethyl acetate, and butanoic acid. The weights on the remaining 31 components were very small, so they were not shown for clarity. Ethanol is produced by yeast or lactic acid bacteria (LAB) metabolism under anaerobic fermentation of soybeans (Park et al., 1994). Acetic acid was also produced during fermentation, and its content correlated with fermentation duration (Namgung et al., 2010). When *t*-test was performed between the peak areas of ethanol from the imported and domestic samples (refer to Figure S2 in the supplementary data), the *p*-value was 0.60, much greater than 0.05. It indicates that the difference is not significant at a confidence interval (CI) of 95 %. The *p*-value resulted from *t*-test of the peak areas of acetic acid (refer to Fig. S2) was 0.20, again larger than 0.05. Therefore, the peak areas of ethanol and acetic acid were not descriptive for the discrimination.

Use of more group-discriminant components instead of all components would be advantageous to improve accuracy by excluding undescriptive and/or redundant components. For this purpose, *t*-tests were performed between the imported and domestic samples using the peak areas of each component, and 5 components yielding *p*-values smaller than 0.05 (indicating that the difference was significant at a CI of 95 %) were chosen. The average peak areas of the selected 5 components and the corresponding standard deviations are summarized in Table S3 (supplementary information).

Fig. 3 (a) through (e) present the peak areas of the 5 *t*-test-selected components in all the imported and domestic samples (refer to the 5 plots inside the black dashed-line box). The corresponding mass spectra are shown in the supplementary data (Figure S3). Average peak areas and standard deviations are also displayed for each component. The *p*-values smaller than 0.05 (95 % CI)), 0.01 (99.0 % CI), 0.001 (99.9 % CI), or 0.0001 (99.99 % CI) are designated by *, **, ***, or ****, respectively. The three components of trichloromethane (Peak 1), 4-methylnonane (Peak 4), and 2,3-pentanedione (Peak 5) yielded *p*-values smaller than 0.0001, signifying a substantial concentration difference between the two groups. The *p*-values for the remaining two components (4-methyldecane (Peak 2) and hexanal (Peak 3)) were smaller than 0.001 and 0.05, respectively.

Trichloromethane (Peak 1, Fig. 3 (a)) can be formed by the reactions between free chlorine and organic matter in foods during the food processing procedure, so the appearances of trichloromethane in various foods such as rice, coffee, tofu, and milk have been reported (Huang & Batterman, 2009). There are several sources of chlorine such as presence of trichloromethane in groundwater, surface water, and drinking water (Zimoch & Łobos, 2016). Vegetation burning, volcanism, and biogenic formation in soil could be responsible for the formation of chloroform, and fungi also could produce chloroform from inorganic chloride (Laturnus et al., 2002). Overall, the origins of the selected components by *t*-test and relative abundance of most of the components between the two sample groups were reasonably explained.

The presences of 4-methyldecane (Peak 2), hexanal (Peak 3), 4-methylnonane (Peak 4), and 2,3-pentanedione (Peak 5) were reported in previous studies (Feng et al., 2017; Joong-Ho et al., 2006; Van Ba et al., 2012). 4-Methyldecane and 4-methylnonane were derived from lipid oxidation during fermentation (Domínguez et al., 2019; Zhang et al.,

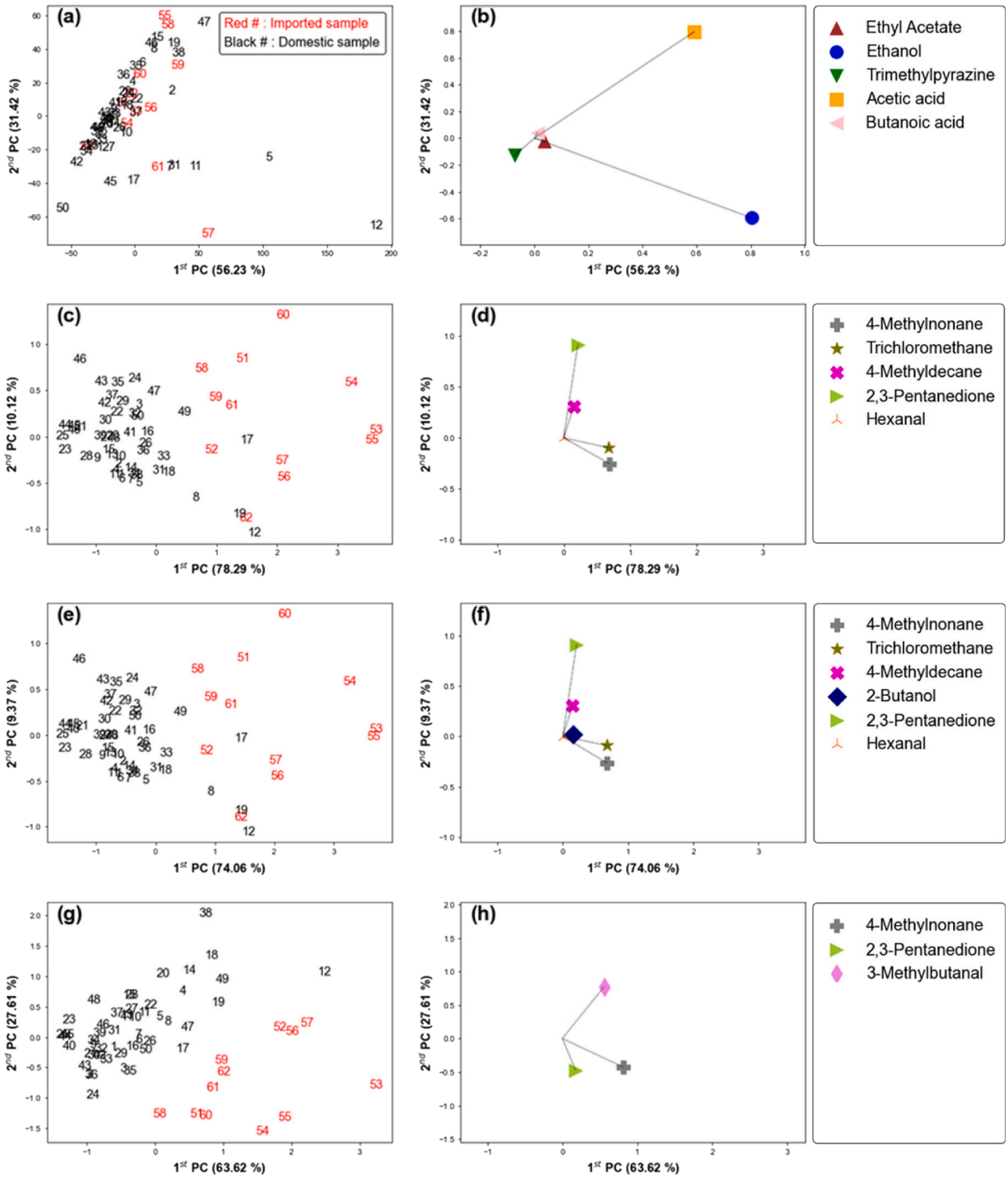


**Fig. 2.** Distribution of the 1st and 2nd PC scores obtained by PCA of all 36 peak areas (a), 5 peak areas selected by *t*-test (c), 6 peak areas selected by VIP score (e), and 3 peak areas selected by IAMB (g). The loadings used to generate the corresponding PC scores are shown to the right of each score scatter plot ((b), (d), (f), and (h), respectively).

2020). Since the fat contents in the imported soybeans were higher than that of the domestic soybeans (NAS, 2017), the concentrations of 4-methylnonane and 4-methyldecane were accordingly higher for the imported samples (refer to Fig. 3 (d) and (b), respectively). Hexanal (Peak 3) can be formed in soybeans from linoleic acid by lipoxygenase (Matoba et al., 1985), although noticeable amounts of hexanal were found only in 2 imported and 2 domestic samples (Fig. 3 (c)). The formation of 2,3-pentanedione (Peak 5) might be associated with an amino acid and a fragment pyruvaldehyde from D-glucose (Yaylayan & Keyhani, 1999).

Table 1 shows the cross-validated discrimination results when the peak areas of the 5 *t*-test-selected components were used. In comparison with the results using peak areas of all 36 components, the accuracies were substantially improved to 95.2–96.2 %, even when employing any

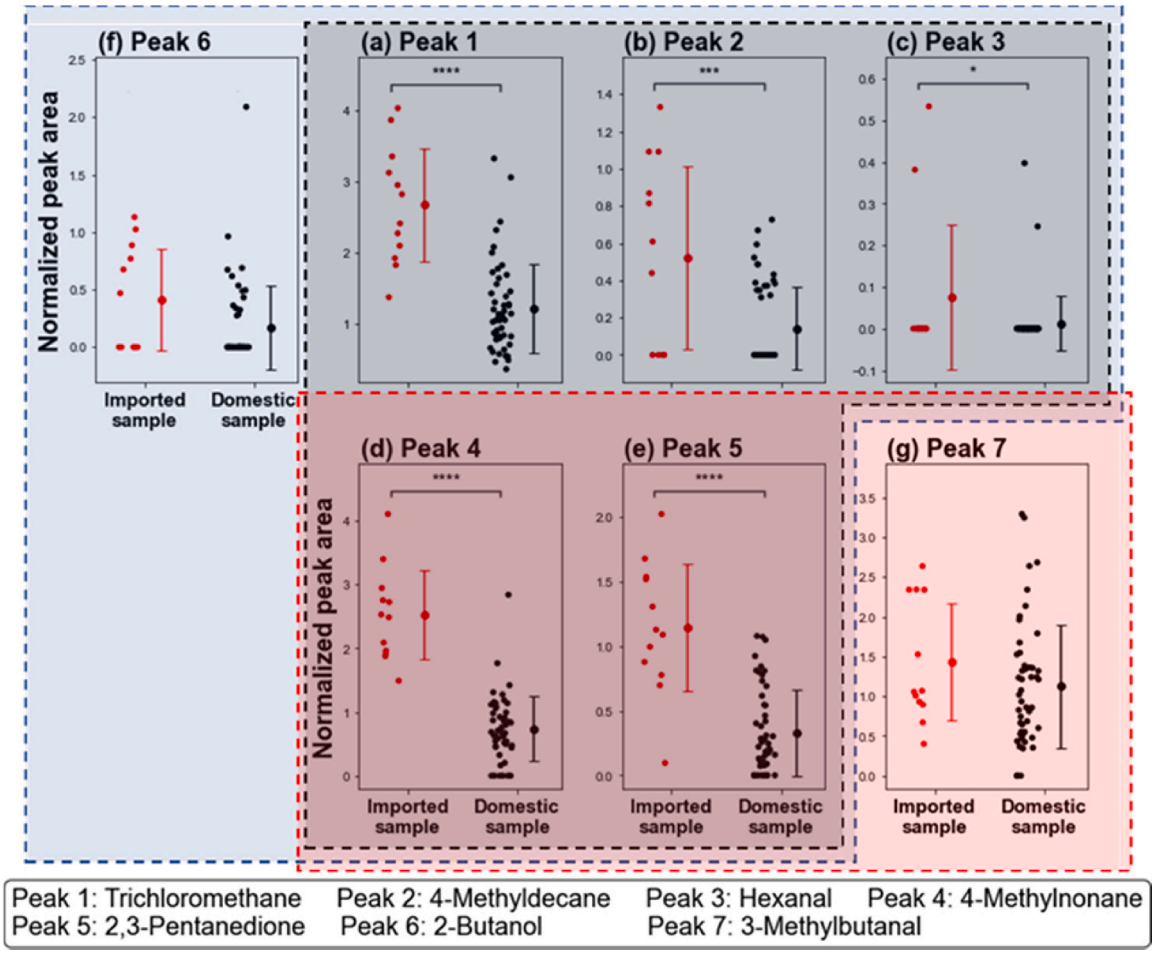


**Fig. 3.** Peak areas of the 5 components selected by *t*-test ((a) through (e) inside the black dashed-line box), 6 components selected by VIP score ((a)~(f) inside the blue dashed-line box), and 3 components selected by IAMB ((d), (e), (g) inside the red dashed-line box). For each component, the average peak areas and corresponding standard deviations are designated. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

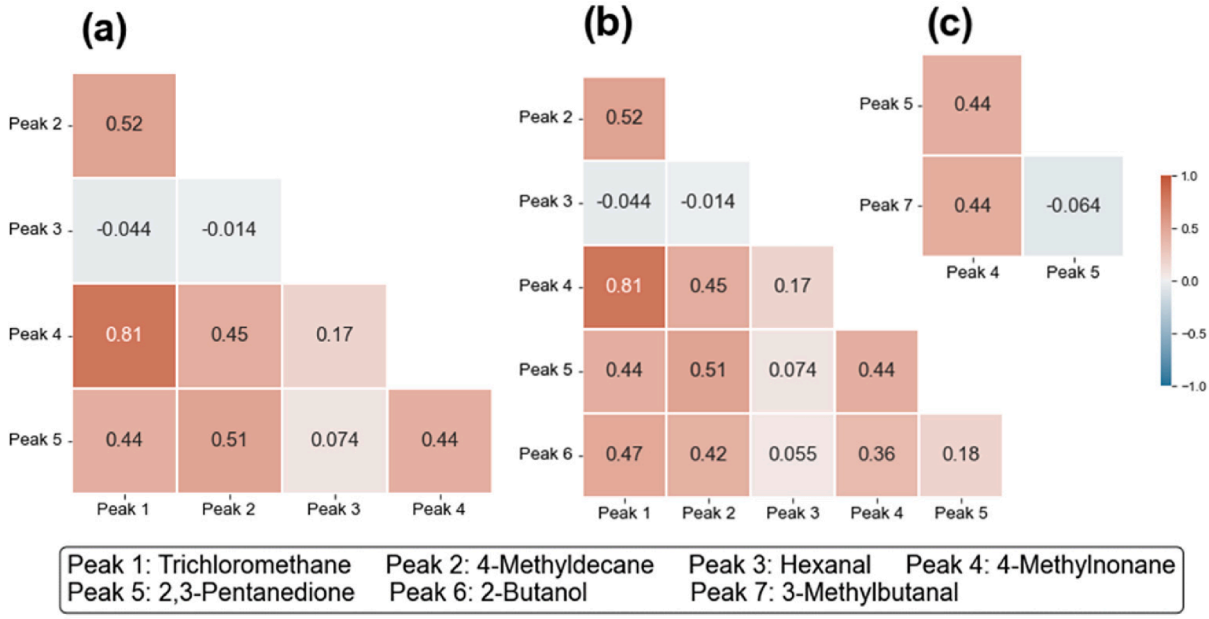


**Fig. 4.** Pearson correlation coefficients (r) among the peak areas of 5 components selected by *t*-test (a), peak areas of 6 components selected by VIP score (b), and peak areas of 3 components selected by IAMB (c). The red and blue colors indicate positive and negative correlations, respectively.

of the three classifiers. Fig. 2 (c) and (d) show the 1st and 2nd PC score distribution and corresponding loadings obtained by PCA of the 5 peak areas, respectively. In the PC score domain, distinction of the imported and domestic samples became clearer, although the #12/17/19 domestic and #52/58/59 imported samples are closely located around the boundary of the two groups. The use of only 5 discriminant components substantially improved the differentiation of two sample groups. The 4 components (trichloromethane, 4-methyldecane, 4-methylnonane, and 2,3-pentanedione) mutually and considerably contributed to the discrimination, as seen in the loading plot.

Next, inter-correlations among the 5 peak areas of the samples were investigated as shown in Fig. 4 (a). The Pearson correlation coefficient (r) between the two compared components is designated. The red and blue colors indicate positive and negative correlations, respectively. In general, it is ideal if the chosen variables are all independent in describing sample composition for discriminant analysis. Variables with considerable inter-correlations would be redundant and better to be excluded. As seen, there was considerable correlation between the peak areas of trichloromethane (Peak 1) and 4-methylnonane (Peak 2) (r: 0.81), while those of the remaining comparisons were low (all the r values below 0.52). When the peak areas of 4 components excluding trichloromethane were adopted for discrimination, the accuracies were 96.4 ± 0.9, 95.5 ± 0.8, and 96.1 ± 1.1 % when using LDA, *k*-NN, and SVM, respectively. The improvement of LDA accuracy (from 95.2 to 96.4 %) by exclusion of trichloromethane was notable. When excluding 4-methylnonane, accuracies based on LDA, *k*-NN, and SVM were 93.9 ± 2.4, 91.8 ± 2.0, and 93.9 ± 1.7 %, respectively. All of these were lower than the respective accuracies using all 5 components. Although the differences of both trichloromethane and 4-methylnonane contents between the two groups were statistically significant, the trichloromethane content seems less descriptive and redundant for discrimination.

Based on a previous publication on the comparative review of feature selection and classification modeling (Azhar and Thomas, 2019), some features are irrelevant to the output of the model and/or redundant. In learning algorithms, these variables introduce noise which degrades the learning performance and makes the prediction degraded. The authors claimed that the removal of these variables was useful for insight into a targeting classification. Therefore, the inferior discrimination accuracy even with employing all the 36 variables was from the inclusion of 31 components yielding the *p*-values greater than 0.05 (not significantly different between the peak areas of the imported and domestic samples), which could be considered as irrelevant variables for the discriminant analysis. Equivalently, it explains the improved accuracy using the 5 *t*-test-selected variables (components).

### 3.3. Discriminant analysis using peak areas of components selected by VIP and IAMB

Among the peak areas of 36 components, those yielding VIP scores greater than 1 were selected. The chosen 6 components are shown inside the blue dashed-line box in Fig. 3. The corresponding mass spectra are shown in the supplementary data (Fig. S2). Their VIP scores ranged from 1.0 to 2.99 (also refer to Table S3). In comparison with the 5 *t*-test-selected components, 2-butanol (Peak 6) was added; the other 5 components were unchanged. The difference in 2-butanol peak area between the two groups (Fig. 3 (f)) was not significant at 95 % CI (*p*-value: 0.056). When the peak areas of the 6 VIP score-selected components were employed for discrimination, the accuracies were slightly inferior (94.7–95.0 %, refer to Table 1) to those using the *t*-test-selected 5 components.

Fig. 2 (e) and (f) display the 1st and 2nd PC score distributions and corresponding loadings obtained by PCA of the 6 selected peak areas by VIP score, respectively. Although the two groups are noticeable, still the #12/17/19 domestic and #52/58/59 imported samples are nearly gathered around the boundary of the two groups. This observation is similar to that in employing the 5 *t*-test-selected components (Fig. 2 (c)). The 4 components (trichloromethane, 4-methyldecane, 4-methylnonane, and 2,3-pentanedione) again considerably contributed to the discrimination, the same as those when using the *t*-test. Trichloromethane and 4-methylnonane with considerable correlation were chosen again, while the correlations in the remaining comparisons were again low (all the r values below 0.49, refer to Fig. 4 (b)). When the peak areas of 5 components excluding trichloromethane were employed, LDA, *k*-NN, and SVM accuracies were 96.4 ± 1.2, 95.5 ± 0.8, and 96.1 ± 1.1 %, respectively, higher than those when using all 6 components. When excluding nonane, LDA, *k*-NN, and SVM accuracies were 94.0 ± 2.6, 91.0 ± 1.9, and 93.0 ± 1.8 %, respectively, which were marginally lower than the corresponding accuracies using all 6 components. Although the VIP score of trichloromethane was high (2.47), its content was not highly descriptive for discriminating the imported and domestic samples.

When IAMB was used, only the peak areas of only 3 components of 4-methylnonane (Peak 4), 2,3-pentanedione (Peak 5), and 3-methylbutanal (Peak 7) were selected (refer to the red dashed-line box in Fig. 3 and the corresponding mass spectra in the supplementary data), and all their inter-correlations were not significant (r values below 0.44, Fig. 4 (c)). 3-Methylbutanal was newly selected, while the other 2 components were chosen by the *t*-test as well as VIP score. It is noteworthy that trichloromethane, the component contributing most to the discrimination in the previous two cases, was not selected by IAMB. Since the correlation between the peak areas of trichloromethane and 4-methylnonane was relatively high as discussed above, the trichloromethane information would be redundant and was excluded, retaining only the 4-methylnonane information for discrimination. The experimental retention index values calculated for 3-methylbutanal, 4-methylnonane, and 2,3-pentanedione were 892, 986, and 1047, respectively, while the reported retention index values are 918, 970, and 1058 as shown in Table S2. The closeness between the two retention index values for each component suggests that the current assignment is reliable.

Although the average peak area of newly chosen 3-methylbutanal is marginally higher for the imported samples (Fig. 3 (g)), their difference between the two groups is not significant (*p*-value: 0.23). 3-Methylbutanal can be derived from leucine, isoleucine, and glucose by Strecker reaction during fermentation (Smit et al., 2009). The average leucine content in domestic soybean was 2.58 wt%, smaller than the 3.32 wt% in American soybeans and the 3.12 wt% in Chinese soybeans (Grieshop & Fahey, 2001; NAS, 2017). The average isoleucine content of domestic soybeans was 1.31 wt%, smaller than those of American and Chinese soybeans (1.91 and 1.72 wt%, respectively) (Grieshop & Fahey, 2001; NAS, 2017). Glucose can be produced from sucrose (Lee et al., 2014), and higher sucrose content in the imported soybean was previously reported (NAS, 2017). These former studies reporting higher contents of leucine, isoleucine, and sucrose in the imported soybean explain the higher 3-methylbutanal content in the imported soybean paste samples in this study.

When the peak areas of 3 IAMB-selected components were used for discrimination, the accuracies became improved, 97.3 and 97.7 % when using *k*-NN and SVM, respectively (Table 1). In the 1st vs· 2nd PC score domain (Fig. 2 (g)), the imported and domestic samples are better discernible without mutual overlaps along the diagonal boundary. Especially, the #12/17/19 domestic and #52/58/59 imported samples that were inseparable in the both PC domains based on *t*-test and VIP score can now be further separable. As shown in Fig. 2 (h), the loadings of the 3 selected components were directed to different positions, and their weights were individually considerable. This observation suggests that the 3 IAMB-selected components are collectively useful for differentiating imported and domestic samples.

When 3-methylbutanal was excluded for discrimination, LDA, *k*-NN, and SVM accuracies were 96.9 ± 0.5, 97.1 ± 0.8, and 97.0 ± 0.9 %, all lower than the respective accuracies using the 3 components. These results demonstrate that the content of 3-methylbutanal was complementary to those of the remaining 2 components and was necessary to

maximize the accuracy further to 97.7 %. As explained earlier, the selection of optimal variables with excluding the redundant variables for the discriminant analysis was responsible for the superior accuracy. Although the number of IAMB-selected components was only three, their peak areas were descriptive to discriminate the imported and domestic samples and mutually independent.

To evaluate the prediction abilities of the models, the classes of external validation samples (5 imported and 10 domestic samples) were determined and resulting accuracies, specificities, and sensitivities were assessed as shown in Table 2. Again, the accuracies in the external validation were most superior with using the 3 IAMB-selected components compared those in the remained cases. The achieved SVM-based accuracy was 95.6 %, slightly lower than 97.7 % (cross-validated accuracy). The marginal decrease of accuracy is probably attributed to the interferences of contained additives in the measurement. As described earlier, among the 5 imported and 10 domestic external validation samples, 3 and 6 samples contained the additives, respectively. So, the influence of additives was relatively larger in the measurement of imported samples, thereby lowering the specificity. Nonetheless, the overall accuracy in the prediction of independent validation was superior, 95.6 %. It supports that the IAMB-selected components found from the no-additive samples (the internal validation set) could be useful for the authentication of various types of soybean paste samples.

**Table 2**
Accuracies, specificities, and sensitivities resulting in the discrimination of imported and domestic soybean paste samples in the external validation set when peak areas of all 36 separated components or selected components among them by *t*-test, VIP score, and IAMB are separately employed for LDA, *k*-NN, and SVM.

| Input | Classifier | Accuracy | Specificity | Sensitivity |
|---|---|---|---|---|
| **Peak areas of all 36 separated components** | LDA | 73.3 ± 4.4 % | 60.7 ± 6.9 % | 79.6 ± 5.6 % |
| | *k*-NN | 61.3 ± 1.9 % | 1.7 ± 3.8 % | 91.1 ± 2.2 % |
| | SVM | 83.0 ± 1.9 % | 49.4 ± 5.4 % | 99.8 ± 0.7 % |
| **Peak areas of 5 components selected by *t*-test** | LDA | 80.6 ± 1.5 % | 42.0 ± 4.6 % | 99.9 ± 0.5 % |
| | *k*-NN | 89.0 ± 1.6 % | 69.9 ± 3.9 % | 98.6 ± 1.2 % |
| | SVM | 84.3 ± 2.7 % | 52.9 ± 8.1 % | 100.0 ± 0.4 % |
| **Peak areas of 6 components selected by VIP score** | LDA | 79.1 ± 1.7 % | 37.2 ± 5.1 % | 100.0 ± 0.3 % |
| | *k*-NN | 89.2 ± 1.9 % | 69.6 ± 4.5 % | 99.1 ± 1.2 % |
| | SVM | 85.6 ± 2.8 % | 57.0 ± 8.4 % | 100.0 ± 0.3 % |
| **Peak areas of 3 components selected by IAMB** | LDA | 91.6 ± 1.2 % | 74.8 ± 3.6 % | 100.0 ± 0.0 % |
| | *k*-NN | 92.9 ± 0.9 % | 78.9 ± 2.5 % | 100.0 ± 0.4 % |
| | SVM | 95.6 ± 1.3 % | 86.9 ± 3.9 % | 100.0 ± 0.0 % |

## 4. Conclusion

The peak areas of 3 components (4-methylnonane, 2,3-pentanedione, and 3-methylbutanal) selected by IAMB were individually descriptive to differentiate imported and domestic soybean paste samples, and the discrimination accuracy was high (97.7 % in the cross-validation. 95.6 % in the external validation), when they were employed for SVM. The exclusion of less descriptive and/or redundant components such as trichloromethane, and inclusion of a more descriptive component such as 3-methylbutanal were responsible for the improved accuracy. The overall results demonstrate the importance of peak (variable) selection among separated chromatographic peaks for reliable and accurate multivariate discriminant analysis.

Since portable HS-GCs are commercially available, at-site identification of the geographical origin of soybean paste is feasible. Reproducibility of measurements may be degraded when a sample is measured using portable instrumentation. Under this situation, components chosen by variable selection methods would be again different from those selected from chromatographic data obtained using well-controlled laboratory instrumentation. Investigation of this interesting issue could be helpful to provide a guideline for real-field application, which is under way in our research group. In addition, similar or upgraded analytical strategies will be explored for country-origin authentication of other odor-rich agricultural products such as onion and garlic. For these studies, the adoption of HS-SPME (solid phase microextraction)-GC-MS is expected to be advantageous to secure more sensitive and accurate authentication.

## CRediT authorship contribution statement

**Seongsoo Jeong:** Investigation, Formal analysis, Visualization, Conceptualization. **Dokyung Kwon:** Methodology, Investigation, Validation. **Jina Lim:** Methodology, Investigation, Validation. **Hanbyeol Jang:** Investigation, Methodology. **Jeongkwon Kim:** Project administration, Writing – original draft, Methodology. **Hoeil Chung:** Project administration, Conceptualization, Writing – original draft, Writing – review & editing, Supervision.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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## Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodres.2023.113492.

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